

1 Solution — GIAM 7.2

Problem 9

1.1 Problem

A *magic hexagon of order 2* would consist of filling-in the numbers from 1 to 7 in a hexagonal array of 7 cells. The magic condition means that each of the 9 “lines” of adjacent hexagons would have the same sum. **Is this possible?**

1.2 Setup

Label the seven cells A, B, C, D, E, F, G as in the figure, with D at the center and the other six cells surrounding it.

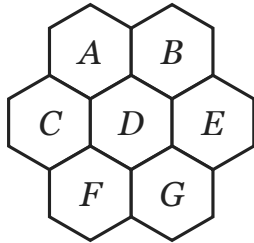


Figure 1.1: The hexagonal array of order 2: seven cells arranged as 2-3-2 rows, with D at the center surrounded by the six outer cells A, B, C, E, F, G .

1.3 Cataloguing the 9 Lines

A “line” is a maximal straight row of adjacent hexagons along one of the three hexagonal axes. Each axis contains 3 lines: a *long* line of 3 cells through the center, plus two flanking *short* lines of 2 cells each.

Axis	Short line	Long line (through D)	Short line
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Horizontal	$\{A, B\}$	$\{C, D, E\}$	$\{F, G\}$
First diagonal	$\{C, F\}$	$\{A, D, G\}$	$\{B, E\}$
Second diagonal	$\{A, C\}$	$\{B, D, F\}$	$\{E, G\}$

Table 1.1

That gives $3 + 6 = 9$ lines:

- **Three length-3 lines** all pass through the center D .
- **Six length-2 lines** sit on the perimeter and avoid D .

1.4 A Useful Cell-Incidence Count

Each cell lies in exactly **3** lines:

Cell	Lines containing it	Count
A	$\{A, B\}, \{A, D, G\}, \{A, C\}$	3
B	$\{A, B\}, \{B, E\}, \{B, D, F\}$	3
C	$\{C, D, E\}, \{C, F\}, \{A, C\}$	3
D	$\{C, D, E\}, \{A, D, G\}, \{B, D, F\}$	3
E	$\{C, D, E\}, \{B, E\}, \{E, G\}$	3
F	$\{F, G\}, \{C, F\}, \{B, D, F\}$	3
G	$\{F, G\}, \{A, D, G\}, \{E, G\}$	3

Table 1.2

The center D sits on all three long lines (and no short lines). Every outer cell sits on 1 long line and **2** short lines. Either way, each cell is on exactly **3** lines.

1.5 Main Argument — A Single Counting Identity

Suppose, for contradiction, that a magic hexagon of order 2 exists, and let S be the common line-sum. Compute the total

$$T = \sum_{\text{line } \ell} \sum_{x \in \ell} (\text{value at cell } x)$$

in two ways.

By lines. There are 9 lines, each summing to S , so

$$T = 9S.$$

By cells. Each cell's value is counted once for every line containing it, i.e., exactly 3 times (by the table above). The sum of cell values is $1 + 2 + \cdots + 7 = 28$. So

$$T = 3 \cdot 28 = 84.$$

Equating the two expressions for T :

$$9S = 84 \implies S = \frac{84}{9} = \frac{28}{3}.$$

But S is a sum of integers (each entry is a positive integer from $\{1, 2, \dots, 7\}$), so S must itself be an integer. Since $28/3 \notin \mathbb{Z}$, this is a contradiction.

Therefore no magic hexagon of order 2 exists. ■

1.6 Alternative Proof — Long Versus Short Lines

The conclusion can also be reached by analyzing the long and short lines separately. Sum over the 3 long lines: every long line passes through D , so D is counted 3 times and each outer cell once.

$$3S = 3D + \sum_{\text{outer}} \text{value} = 3D + (28 - D) = 28 + 2D.$$

Sum over the 6 short lines: no short line touches D , and each outer cell sits on exactly 2 short lines.

$$6S = 2 \sum_{\text{outer}} \text{value} = 2(28 - D).$$

Dividing the second by 2:

$$3S = 28 - D.$$

Both expressions equal $3S$, so $28 + 2D = 28 - D$, hence $3D = 0$, hence $D = 0$. Impossible, since $D \in \{1, 2, \dots, 7\}$. ■

(The first proof's $S = 28/3$ comes out as the *average* of these two expressions: $S = (28 + 2D)/3$ at $D = 0$ gives $28/3$, and indeed $9 \cdot 28/3 = 84 = 3 \cdot 28$.)

1.7 Why the Obstruction Is Robust

The argument hinges on three numerical facts:

1. The sum of all entries is 28.
2. Each cell sits on exactly 3 lines.
3. There are exactly 9 lines.

The contradiction comes from $9 \nmid 3 \cdot 28$. Equivalently, $28 \not\equiv 0 \pmod{3}$ (since $9 = 3 \cdot 3$ and the factor of 3 in the numerator from cell-incidence is absorbed). Any alternative set of 7 distinct integers with the same property “fills in” would face the same divisibility obstruction unless its sum is divisible by 3.

For the set $\{1, \dots, 7\}$: $28 \equiv 1 \pmod{3}$, and $28 \not\equiv 0 \pmod{3}$. So no magic hexagon.

1.8 Remarks — Magic Hexagons of Other Orders

A magic hexagon of order n has $N = 3n^2 - 3n + 1$ cells arranged in $2n - 1$ horizontal rows, with line-lengths varying from n (at the edges) to $2n - 1$ (through the center) and back to n . There are $3(2n - 1)$ lines in total, and each cell still sits on exactly 3 lines (one per axis).

By the same cell-incidence argument as above:

$$3(2n-1) \cdot S = 3 \cdot \frac{N(N+1)}{2}, \quad \text{so} \quad S = \frac{N(N+1)}{2(2n-1)}.$$

For S to be an integer is a *necessary* condition. The table for small n :

n	N	$N(N+1)$	$2(2n-1)$	S (if integer)	Magic hexagon exists?
1	1	2	2	1	trivially (1 cell)
2	7	56	6	$28/3$ (not integer)	No — this problem
3	19	380	10	38	Yes, and unique (Adams, 1910)
4	37	1406	14	$1406/14$ (not integer)	No
5	61	3782	18	not integer	No

Table 1.3

So integer-arithmetic alone rules out $n = 2, 4, 5, 6, \dots$ for most cases (one has to chase the divisibility carefully). The order-**3** magic hexagon, discovered by C. W. Adams in 1910, is the *unique* nontrivial magic hexagon up to rotation and reflection — each of its 15 lines sums to 38.

1.9 A Curious Fact

The arithmetic identity $9S = 3 \cdot 28$ in our proof can be read as: *if* a magic hexagon of order 2 existed, the magic constant would be three times the average cell value $\bar{c} = 28/7 = 4$:

$$S = \frac{3 \cdot \text{sum}}{9} = \frac{\text{sum}}{3} = 3\bar{c} \cdot \frac{7}{9}.$$

This is the natural prediction since each line is a length-2 or length-3 “sample” of the cells. The trouble is that the *integer count* doesn’t quite cooperate: $9 \nmid 3 \cdot 28$. By contrast, for the order-3 case, $45 \mid 3 \cdot 190 = 570$, giving $S = 570/45 = 38$ neatly.